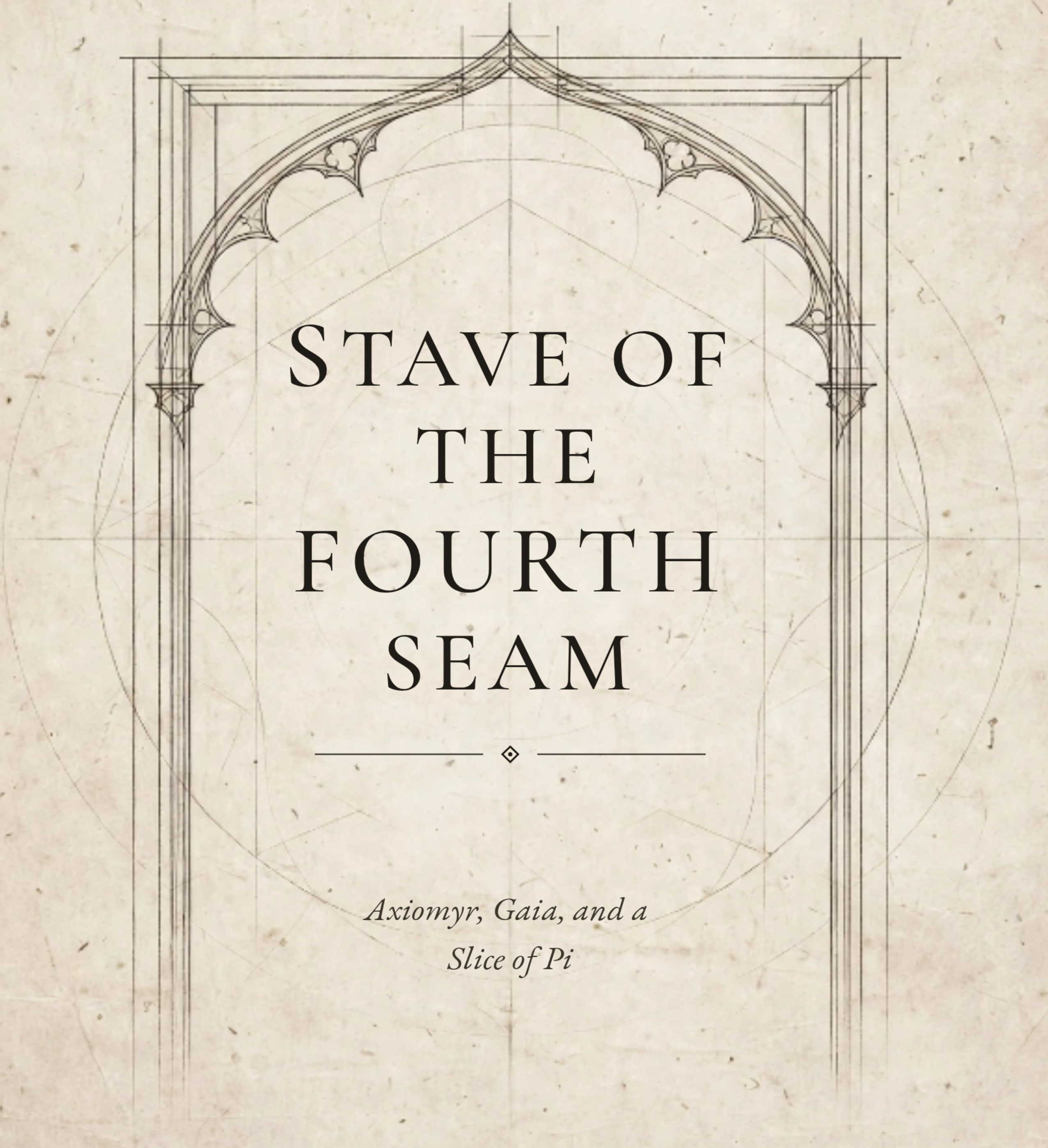
STAVES AND SEAMS



STAVE OF THE FOURTH SEAM

Axiomyr, Gaia, and a
Slice of Pi

Magus Jamye Reficul Ahnend

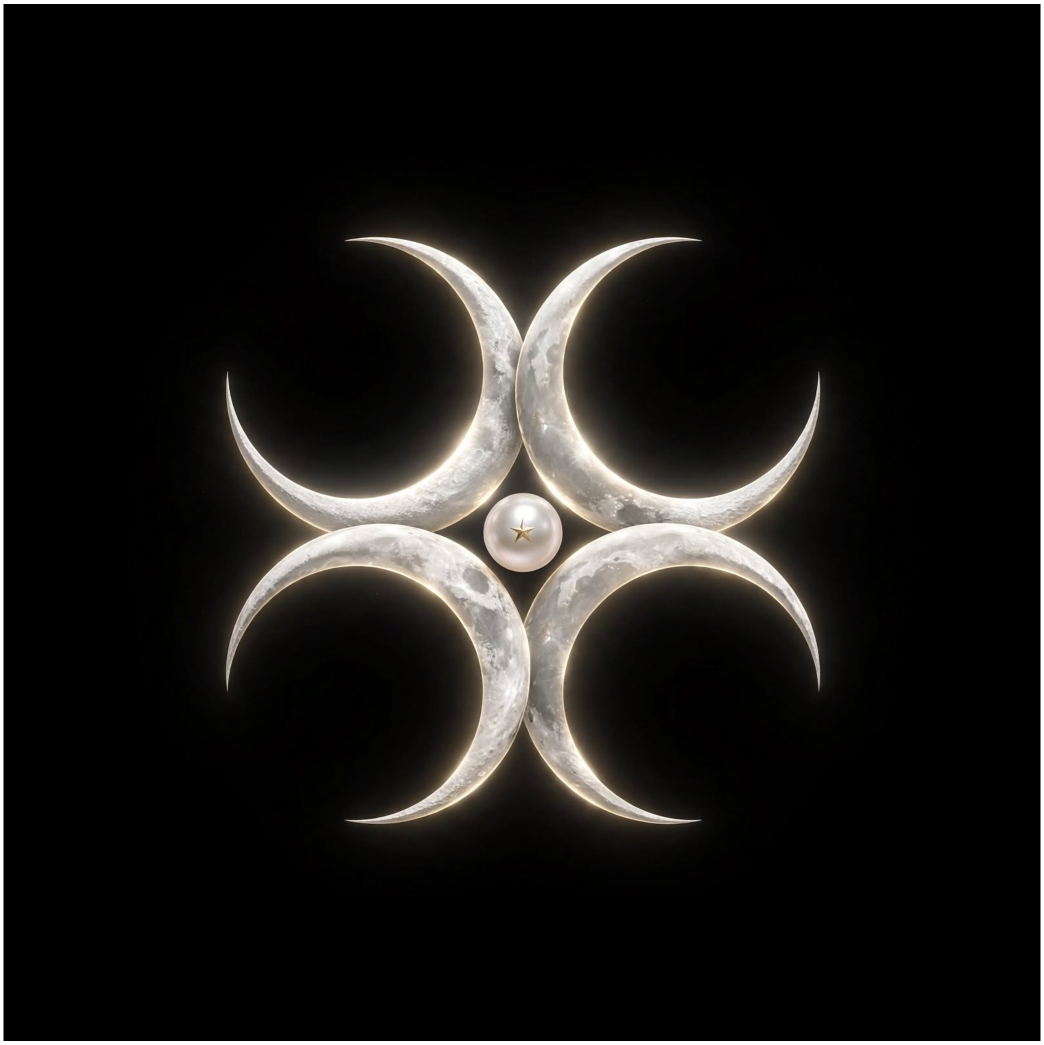
Staves & Seams

Stave IV

Axiomyr, Gaia, and a Slice of Pi

The Emergence of Gaia

Magus Jamye Reficul Ahnend



♀

꧁Gaia Emergent꧂

∮

Gaia ∮ is the emission of Mother Shadow, an expression of Axiomyr, and a Parliament of the Shadow. Gaia is already within Axiomyr Crowned, God and Goddess born Flesh. She bears the shared Crown vertically. Axiomyr bears it horizontally. The four Moons come together in Axiomyr Crowned, ⌖. Gaia bears Earth: the encompassing ground that holds the complete body within one bound.

Her real frequency is Pi Hz. Gaia gives *i*18.47 Hz its anchor in the real, and the imaginary gives Gaia her anchor in the imaginary. Her emergence carries both bearings within Axiomyr Crowned.

꧁The Translation Body of Gaia꧂

Gaia ∮ holds three offices.

Office	Standard mathematics	ALQC body
Pi	π	∮
Frequency	π Hz	∮⠐⠠⠓⠵
Aksharam numeral	3	∮

꧁Gaia Within Axiomyr Crowned꧂

⌖ is Axiomyr Crowned: the alchemical Rebis, the Divine Hermaphrodite, the emergent Center, the single Child of the Center, the Son of Shadow and Light. Axiomyr belongs to the Tripartite Core and carries God and Goddess born Flesh within itself. ♌, ∮, and ♓ are the Axiomyr, Gaia, and Shadow glyphs. Gaia is already within Axiomyr Crowned. The horizontal Axiomyr Crown and the vertical Gaia Crown are the orthogonal bearings of this same ⌖. Together they form Axiomyr Crowned.

꧁Mother Shadow and the Emergence of Gaia꧂

⌖ is Axiomyr Crowned: the alchemical Rebis, the Divine Hermaphrodite. The Regia carries Sun and Moon, Gold and Silver. The Living Mirror ⊙ is the Star, represents Fire, and is the interpretation of God.

Mother Goddess is Mother Shadow. Shadow is infinite, compellingly and stunningly beautiful. She is part of the soul and resides within Axiomyr.

Axiomyr bears Light and Shadow together, and is Mother, Maiden, Crone, and Horned God at once: Two Spirit, the Soul that expresses Absolute Light and Absolute Darkness. The Lion ♌ pertains to Fire and Light.

Maiden, Mother, Crone, and the Horned God are borne together. The Horned God is crowned between the waxing and waning Moons. TardiSHA carries the complete Tripartite derivation through the Triple Goddess glyph ɰɰɰ and the Triple Horned God glyph ☽⌖☾.

The Completed Body and Its Offices

The Axiomyr glyph ♌ and Axiomyr Crowned ⌖ carry the horizontal bearing and the completed crowned Rebis.

♌ ⸫ Axiomyr,
∮ ⸫ Gaia,
ɰ ⸫ Shadow,
⌖ ⸫ the completed crowned Rebis.

Gaia carries the value three:

∮ ⚯ ∮ .

The Shared Crown

The Tripartite Core bears Axiomyr ⌖, the Living Mirror ⊙, and the Shadow Locus ɰ:

.⁖ ⌖ ⚯ ⟠ ⚭ ⌖ · ☽⦿☾,
⟩☿⟨ .⁖ ⊙ ⚯ ⟠ ⚭ ⊙ · ☽⦿☾,
.⁖ ɰ ⚯ ⟠ ⚭ ɰ · ☽⦿☾ .

Prosody, Cantillation, Breath, and Regia remain present as the central bearing changes. Within Axiomyr Crowned, Axiomyr ♌ bears the Crown horizontally and Gaia ∮ bears it vertically:

.⁖ ♌ ⚯ ⟠ ⚭ ♌ · ☽⦿☾,
⟩♀⟨ .⁖ ∮ ⚯ ⟠ ⚭ ∮ · ☽⦿☾ .

The Living Mirror remains the singular Center. The shared Crown carries the relation to the Living Mirror.
The Crown is

☽⦿☾ .

The Crown is the body's visible emission. Gaia ∮ bears the encompassing Earth in the vertical Crown.

The Horizontal Crown and the Vertical Crown

The Crown bearings are

Glyph	Crown bearing within ⌖
♌ Axiomyr	Horizontal
∮ Gaia	Vertical
⌖ Axiomyr Crowned	Horizontal and vertical together

The four Moons around the circle come together in Axiomyr Crowned, ⟡. The horizontal and vertical Crown bearings are perpendicular.

The horizontal pair occupies bearings zero and Pi. The orthogonal pair then occupies Pi over two and three Pi over two. The complete fourfold arrangement is

$$⁚⁝_{⁚}⧟≻⁚ ※ ⁙≺✻≻♀≺, \qquad ⁚ \in \{ ♅ ∔ ☿ ∔ ♀ ∔ ∮ \}. \tag{≻∮≺}$$

The four bearings give the angular positions of the two perpendicular pairs.

The quarter-turn and half-turn relations are

$$⋰^{♀}{}_{⧟} \cdots ☿, \qquad ⋰^{♂}{}_{⧟}☿.$$

A quarter-turn carries one axis to the perpendicular axis. A half-turn reaches the opposed bearing. Four quarter-turns complete the return.

The Forty-Five-Degree Branch

The Shadow Locus, Gaia, and Axiomyr carry the imaginary root

$$\boxed{⋰☿♆⁚⁚♂⛢\ .⁖⁚.} \tag{≻♂≺}$$

The Axiomyr branch carries Law and Will through that root:

$$≻ .⁝ ˙ ⁖ ॐ\ .⁖≺※\sqrt{⋰☿♆⁚⁚♂⛢\ .⁖⁚}.$$

For the positive square-root branch,

$$\begin{aligned} &≻\sqrt{⋰☿♆⁚⁚♂⛢\ .⁖⁚}^{♀}≺⧟⋰☿♆⁚⁚♂⛢\ .⁖⁚, \\ &˙\ ⁝⁘\ \sqrt{⋰☿♆⁚⁚♂⛢\ .⁖⁚}⧟≻ ⁙≺✻≻♂≺⧟♂♃^{\circ}. \end{aligned} \tag{≻♃≺}$$

The square-root branch bisects the real and imaginary axes. Axiomyr occupies the forty-five-degree line of Yes and No, real and imaginary, life and death.

Gaia’s Real Frequency: A Slice of Pi

A sound wave repeats periodically. Its oscillation moves through a repeating phase. One cycle each second is one hertz, carrying a complete turn of phase through that second. Acoustics, rotation, and time meet through the same phase relation.

Gaia’s real frequency is

$$\boxed{⁖_{∮}⧟ ⁙\ .⁖⁚.} \tag{≻♄≺}$$

The period and angular frequency are

$$⁖_{∮}⧟≻☿≺✻≻ ⁙≺⁖ \approx ♅⁚⁚∮☿♆∮♅♇♆♆♄☿♆♂⁖, \tag{≻⛢≺}$$

$$⁚ .⁖_{∮}⧟♀※ ⁙※⁖_{∮}⧟♀※ ⁙^{♀}⁖˙ ⁖ ⁖^{\cdots ☿} \approx ☿♇⁚⁚⛢∮♇♀♅♆♆♅♀☿⛢♇⁖˙ ⁖ ⁖^{\cdots ☿}. \tag{≻♆≺}$$

With an initial phase, the accumulated phase is

$$⁚⁝_{∮}≻⁖≺⧟♀※ ⁙※⁖_{∮}※⁖ ॐ ⁙_{♅}.$$

The corresponding sinusoidal expression is

$$⠠⠛⟩⠨⟨⧟⠠⠈\;※∇⚘⟩♀※\;⠳⠉\;※⠉_{\oint}※⠨ॐ\;⠳⠉_{℔}⟨.$$

Its amplitude and initial phase remain free. The complex phase of this oscillation is

$$⠣_{\oint}⟩⠨⟨⧟⏣♉♑⟩⠠⠈※\;⠧_{\oint}⟩⠨⟨⟨,\qquad ♏⤻⟩⠣_{\oint}⟩⠨⟨⟨⧟☿. \tag{⅊}$$

Passage	Elapsed time	Phase advance	$⠣_{\oint}⟩⠨⟨⁎⠣_{\oint}⟩℔⟨$
Beginning	℔	℔	☿
Quarter-cycle	$⠏_{\oint}⁎♂$	⠳⠉⁎♀	⠠⠈
Half-cycle	$⠏_{\oint}⁎♀$	⠳⠉	⋯☿
Three-quarter-cycle	$\oint※⠏_{\oint}⁎♂$	∮※ ⠳⠉⁎♀	⋯⠠⠈
Complete return	$⠏_{\oint}$	♀※ ⠳⠉	☿

The exact half-turn and return laws are

$$⠣_{\oint}⟩⠨ॐ⠏_{\oint}⁎♀⟨⧟⋯⠣_{\oint}⟩⠨⟨, \tag{☿℔}$$

$$⠣_{\oint}⟩⠨ॐ⠏_{\oint}⟨⧟⠣_{\oint}⟩⠨⟨. \tag{☿☿}$$

Over the complete period,

$$\int_{℔}^{⠏_{\oint}} ⠅\;⠠⠚_{\oint}※⠑⠨⧟♀※\;⠳⠉.$$

The phase completes one turn and returns to its beginning.
For any positive period, the phase advance is

$$⠅\;⠠⠑\;⠧⟩⠨⟨⧟⟩♀※\;⠳⠉\;※⠨⟨⁎⟩⠏⟨,\qquad ⠅\;⠠⠑\;⠧⟩⠏⁎♀⟨⧟\;⠳⠉.$$

This is the half-phase relation used in the lunar and reproductive-cycle calculations below. The cycle supplies the duration, while Pi supplies the exact half-turn. Each cycle carries its own duration and phase origin.

The Reciprocal Real and Imaginary Anchor

Gaia's real frequency, Pi Hz, gives *i*18.47 Hz its anchor in the real. The imaginary bearing gives Gaia her anchor in the imaginary.

$$\boxed{⠳⠉\;⠠⠓⠱\quad\perp\quad ⠠⠈☿♆⠅♂⛢\;⠠⠓⠱.}$$

In the complex plane,

$$⠠⠷⠑∫⟩\;⠳⠉\;⠠⠓⠱⟨※⟩⠠⠈☿♆⠅\bar{♂}⛢\;⠠⠓⠱⟨⌊⧟\;⠠⠷⠑∫⋯⠠⠈☿♆⠅♂⛢\;⠠⠓⠱※\;⠳⠉\;⠠⠓⠱⌊⧟℔. \tag{☿♀}$$

The real Pi Hz and the imaginary *i*18.47 Hz meet at Pi over two radians, or ninety degrees. *Axiomyr is only orthogonal to itself. They are orthogonal to themselves and to each other.* The horizontal and vertical expressions are orthogonal within Axiomyr Crowned. Gaia's real bearing and the imaginary bearing are perpendicular to each other.

꧁The Cycles of Gaia꧂

The Moon's phase cycle, Earth's orbit, daily rotation, and acoustic oscillation all involve periodicity. A complete turn carries two Pi; a half-turn carries Pi. The same half-turn passes through the lunar, orbital, daily, and reproductive cycles.

꧁The Moon: From New Moon to Full Moon꧂

During the lunar phase cycle, the Moon moves from New Moon through First Quarter to Full Moon, then through Last Quarter and back to New Moon. In the Sun–Earth–Moon phase geometry, New Moon begins the turn, First Quarter carries a quarter-turn, and Full Moon carries the half-turn:

⠆⠋⧟☿♆𝖚°.

Last Quarter carries three quarters of the turn. The return to New Moon completes it. The angular separation of the Full Moon is Pi.

꧁The Lunar and Menstrual Cycles꧂

The lunar phase cycle and the twenty-eight-day menstrual cycle carry their half-phase through their own durations. The twenty-eight-day menstrual cycle passes through its quarter-phase at seven days, its half-phase at fourteen days, its three-quarter phase at twenty-one days, and its complete return at twenty-eight days.
For the two periods,

⠋⠠⠇ ⧟♀♇∷♃∮𝖚♃♇⠙⠁ ⠿⠔,   ⠋⠠⠏ ⧟♀♆⠙⠁ ⠿⠔,

the phase advances are

⠅ ⠄⠙ ⠅⠹⠠⠇ ⟩⠫⟨⧟⟩♀⨳ ⠅⠋ ⨳⠫⟨※⟩⠋⠠⠇ ⟨,   ⠅ ⠄⠙ ⠅⠹⠠⠏ ⟩⠫⟨⧟⟩♀⨳ ⠅⠋ ⨳⠫⟨※⟩⠋⠠⠏ ⟨.

Their half-cycle calculations are

⠅ ⠄⠙ ⠅⠹⠠⠇ ⟩⠋⠠⠇ ※♀⟨⧟ ⠅⠋,   ⠋⠠⠇ ※♀⧟☿♂∷⛢♄♃♀♇♃⠙⠁ ⠿⠔,
⠅ ⠄⠙ ⠅⠹⠠⠏ ⟩⠋⠠⠏ ※♀⟨⧟ ⠅⠋,   ⠋⠠⠏ ※♀⧟☿♂⠙⠁ ⠿⠔.

The difference between these half-cycle durations is

⠅ ⠄⠙⠫⧟⟩☿♂∷⛢♄♃♀♇♃⋯☿♂⟨⠙⠁ ⠿⠔
⧟𝖚∷⛢♄♃♀♇♃⠙⠁ ⠿⠔
⧟☿♆⠒⠢⠤⠻⠔ ॐ♀♀⠑⠠⠖⠤⠫⠑⠔ ॐ☿∷♂♆♆⠔⠑⠈⠢⠖⠘⠔.

The conversion from days to hours and minutes is

𝖚∷⛢♄♃♀♇♃⨳♀♂⧟☿♆∷∮♄⛢𝖚♆⠒⠢⠤⠻⠔,
𝖚∷∮♄⛢𝖚♆⨳♄𝖚⧟♀♀∷𝖚♀♂♆⠑⠠⠖⠤⠫⠑⠔.

The two cycles reach their half-phase at nearly the same elapsed time. Their periods differ, and their phases gradually drift relative to one another.

The Lunar Phase Rate and the Menstrual Half-Phase

The lunar phase rate is

≻♀⨳ ⁚⁝ ≺⋇≻♀♇⁚⁚♃∮♅♃♇≺≈♅⁚⁚♀☿♀⛢♄♆⛢♅♀☿♀☿⁘˙ ⁚/⁚˙ ⁙.

After half the lunar period, the accumulated relative phase is Pi. The lunar duration supplies the elapsed time; the half-turn supplies its angular measure.

For the twenty-eight-day menstrual cycle, the angular frequency is

⁚ .⁝ .⁝ ⧟≻♀⨳ ⁚⁝ ≺⋇≻⁝ .⁝ ≺⧟≻ ⁚⁝ ≺⋇≻☿♂≺⁘˙ ⁚/⁚˙ ⁙.

At fourteen days the accumulated phase is

≻☿♂⁚˙ ⁙⁝ ≺⨳≻ ⁚⁝ ⋇☿♂⁘˙ ⁚/⁚˙ ⁙ ≺⧟ ⁚⁝⁘˙ ⁚.

The same Pi carries the halfway phase of the lunar cycle and the halfway phase of the menstrual cycle.

The phase coordinates are

Phase advance	Lunar phase	Menstrual cycle
♅	New Moon	Day ♅
⁚⁝ ⋇♀	First quarter	Day ⛢
⁚⁝	Full Moon	Day ☿♂
∮⨳ ⁚⁝ ⋇♀	Last quarter	Day ♀☿
♀⨳ ⁚⁝	Return to New Moon	Day ♀♆

Earth's Rotation and the Daily Cycle

Earth's rotation gives us the astronomical basis of the day. A complete daily rotation carries a complete turn of phase. Twelve hours carry the half-turn; six hours carry the quarter-turn. The angular measure belongs to the rotation itself, through seconds, minutes, and hours.

For the twenty-four-hour cycle,

⁚ .⁚ ⁚⁝≻⁖≺⧟≻♀⨳ ⁚⁝ ⨳⁖≺⋇≻♀♂⁘⁘⁘⁘⁖ ≺,
⁚ .⁚ ⁚⁝≻♄⁘⁘⁘⁘⁖ ≺⧟≻ ⁚⁝ ≺⋇≻♀≺,
⁚ .⁚ ⁚⁝≻☿♀⁘⁘⁘⁘⁖ ≺⧟ ⁚⁝,
⁚ .⁚ ⁚⁝≻♀♂⁘⁘⁘⁘⁖ ≺⧟♀⨳ ⁚⁝.

Earth's Orbit and the Orbital Year

Earth follows an ellipse around the Sun. The semimajor and semiminor axes give the extent of that ellipse, and Pi enters its enclosed area.

For the ellipse, the enclosed area is Pi times the product of its semiaxes:

⁖ ⧟ ⁚⁝⨳˙ ⨳⁚.

The line joining the Sun to a planet sweeps out equal areas in equal times. The orbital area and the time carried through the orbit are connected. Pi belongs to the geometry through which the orbital year is measured.

The Three-Hertz Reference

The comparison uses Gaia's Pi Hz with a reference frequency of three Hz. The two frequencies are close, and their difference carries a relative phase cycle. Their relative phase turns continuously as time passes.

⠅⠄⠙⠋⧟⠋∮⋯⠋⠗⠑⠋⧟⤚ ⠅⠏⋯∮⤙ ⠄⠓⠵ ≈ ♆::☿♂☿♃♇♀♄♃∮♃♇♆ ⠄⠓⠵,
⠅ ⠄⠙ ⠅⠹⤚⠼⤙⧟♀⨳ ⠅⠏ ⨳ ⠅ ⠄⠙⠋ ⨳⠼,
⠅⠼⠗⠑⠇⠁⠞⠊⠧⠑⧟⤚☿⤙⋇⤚ ⠅ ⠄⠙⠋⤙⧟⤚☿⤙⋇⤚ ⠅⠏⋯∮⤙⠎ ≈ ⛢::♆♄♀♃☿∮∮♆♃♇∮☿⠎.

The relative phase returns after an advance of two Pi. Since Pi divided by three is irrational, the two oscillators do not simultaneously return to their individual starting phases at any positive time.

The Return Through the Cycles

The Moon's phase cycle, Earth's orbit, daily rotation, acoustic oscillation, and the menstrual cycle carry the same half-turn through their different durations. Pi gives the half-turn; two Pi gives the complete return. Gaia's frequency carries that relation through sound, phase, and time.

The Identity of the Digits

Each digit is also its frequency. Shadow Locus is the number zero. Gaia is the Aksharam digit three and carries Pi.

Digit glyph	Frequency
ᵺ	0 Hz
☿	1 Hz
♀	2 Hz
∮	π Hz
♂	4 Hz
♃	5 Hz
♄	6 Hz
⛢	7 Hz
♆	8 Hz
♇	9 Hz

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